

Alfido Tech Internship – web traffic

Website Traffic Analysis Report

Objective

The objective of this project is to analyze website traffic data to understand visitor activity, identify traffic patterns, and gain insights into user behavior. The analysis was performed using Python in Jupyter Notebook with the Pandas and Matplotlib libraries.

Dataset Overview

The dataset contains website event records with the following information:

- Event
- Date
- Country
- City
- Artist
- Album
- Track
- ISRC
- Link ID

The data was examined to understand website activity across different countries and time periods.

Summary Statistics

The dataset was analyzed by:

- Calculating the **Mean, Median, and Mode** (where applicable).
- Checking for **Missing Values**.
- Identifying **Data Types** of all columns.
- Reviewing the overall structure and quality of the dataset before visualization.

Visualizations

The following visualizations were created:

1. **Bar Chart:** Top 10 Countries by Website Events.
2. **Line Chart:** Website Events Over Time.

3. **Pie Chart:** Distribution of Website Events.

These charts help identify the major traffic sources, event trends, and the geographical distribution of website activity.

Executive Summary

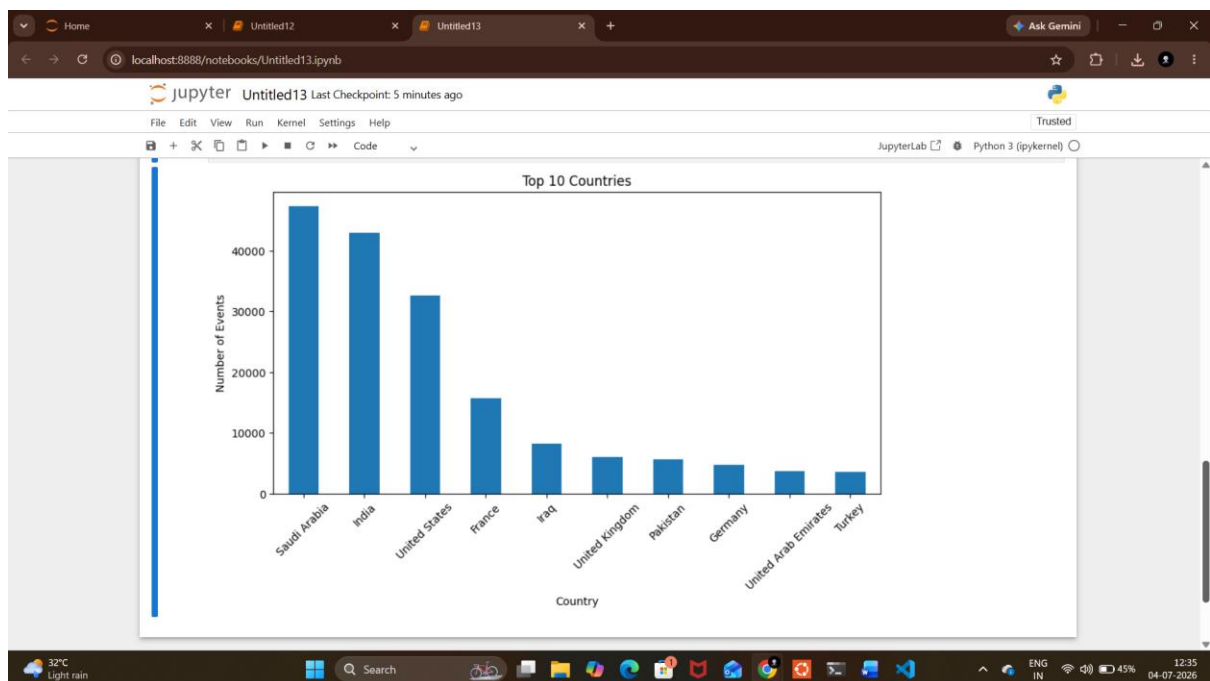
The website traffic analysis provides insights into visitor interactions recorded through website events. The analysis identified the countries generating the highest traffic and showed how website activity changes over time. Event distribution revealed the most common user interactions. The dataset was also checked for missing values and correct data types to ensure reliable analysis. Overall, the results help understand user engagement and support data-driven business decisions.

Actionable Insights

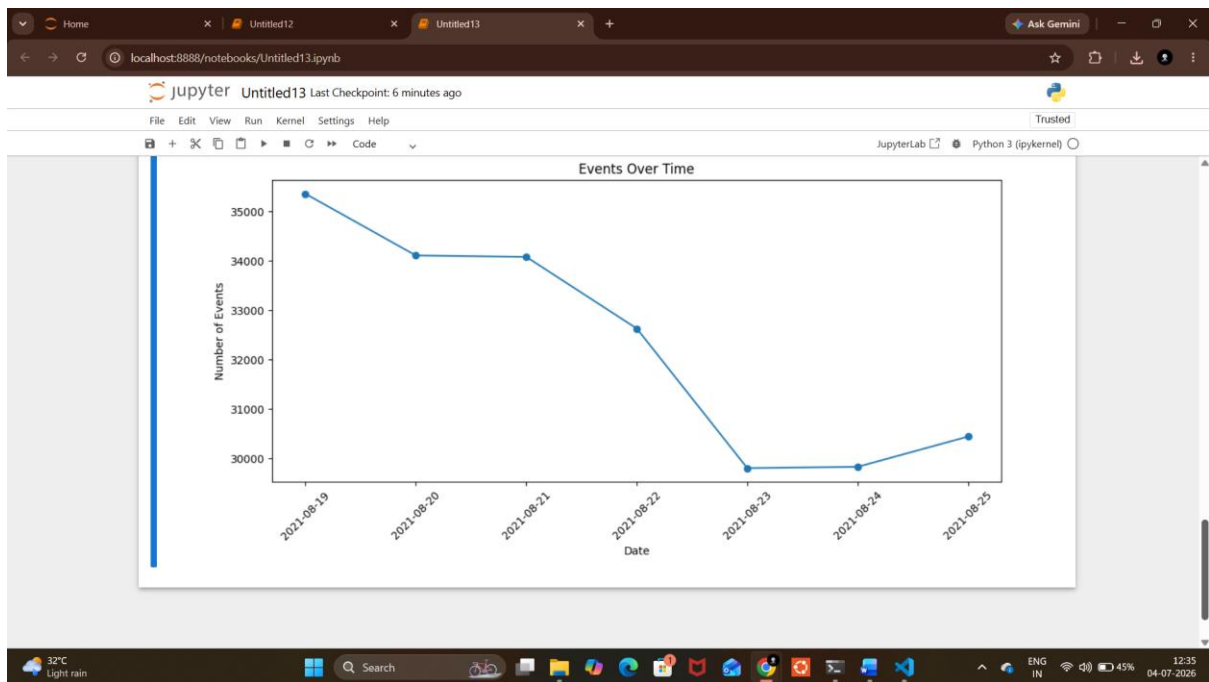
1. **Focus on High-Traffic Countries:** Allocate marketing resources to countries that generate the highest number of website events.
2. **Optimize Content Based on Traffic Trends:** Publish important content or promotional campaigns during periods with higher visitor activity.
3. **Improve User Engagement:** Analyze the most common event types to enhance website navigation and increase user interaction.

Visualisation:

1. Bi chart:



2:Histogram:



3:Pie chart:

